

Pulsed, Not Pressed: State-Space Injection Reaches Parity with Text Interruption in Open-Ended Generation

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Abstract

A research program on machine novelty has intervened at three depths of a language model — the token distribution, the context, and the weights — and certified the limit of each. This paper opens the fourth: the residual stream during live generation. We introduce a rank-1/2 operator algebra on hidden states (translations toward model-derived concept directions, closed-form rotors, projections, repulsion from a moving mean; an anchor and a norm-shell as geometric guards; doses in units of the premise-state norm, which transfer across models), and evaluate it under a deception-resistant protocol: judged surprise, connection and coherence on generated-only windows, plus an exact program verifier. Five pre-registered findings. (i) *The temporal law*: with direction, layer and per-token dose fixed, continuous injection lowers judged surprise (dose-response, $p = 0.81$ against, coherence collapsing), while the same directions pulsed for 32 tokens every 300 raise it (+0.90, $p = 0.006$). (ii) Geometric guards do not repair damage; they license dose — and at guarded dose 2.0 the pulse reaches statistical *parity* with the best prompt-side intervention we know, text interruption (3.40 vs 3.64, two-sided $p = 0.53$; connection identical), without spending a token of the stream. (iii) The result replicates in-family at 14B with an exact tie (3.48 vs 3.48; replication $p = 0.002$). (iv) Mechanistically, habituated generation carries a *silent schematic orbit* — matched token-state periodicity without verbatim repetition — and both winning interventions flatten the internal spectrum. (v) On verified code search the same operators buy no tail depth at all: best distances to classical baselines are identical across arms, dissociating judged experience from discovered value. We also map the boundaries: late layers act as token bias, early layers destabilize into register basins, aligned models break under forced continuation, and repulsion from a moving reference is self-limiting. We release the instrument, every cell, and the dated pre-registrations.

1 Introduction

Can a language model produce something genuinely new, and which interventions actually help? A sequence of studies has approached this question one substrate at a time, under a fixed measurement discipline: judged surprise, connection and coherence on windows containing only generated text (so that the judge can never be shown the intervention itself) [11], and, in verifiable domains, an exact evaluator with classical baselines [12]. At the *token* level, anti-probable sampling bought only surface novelty. At the *context* level, periodically interrupting a habituated stream with a new subject raised judged surprise from 1.8 to ~ 3.0 — the flagship positive of the program — while constant pressure (a continuity connective) was worse than nothing [11]. At the *weight* level, consolidating a model’s own verified findings moved the mean toward the known good and never beyond it, paying for mass with tails [12]; in verified search, operator packages composed additively and stalled at a construction-family plateau (companion paper, in moderation at the time of writing). Between the context and the token sits the object none of these touched: the residual stream $h_\ell(t)$ — the thought in progress. Recent work proves that steered activations reach states that no discrete prompt reaches [10], but leaves their *value*

untested; activation steering itself is an industrial control tool — removing behaviors, enforcing styles [1, 6, 13, 16, 18] — that has not been pointed at creation under a deception-resistant measure.

This paper does that. Its contributions:

1. **An instrument** (§2): a rank-1/2 operator algebra on the residual stream (§2) with model-derived directions, geometric guards, and a dimensionless dose unit that transfers across models; dual MLX/PyTorch backends, bit-identical on the algebra.
2. **The temporal law** (§3): continuous injection has no fertile band — more dose, less judged surprise — while the same vectors, pulsed, are fertile. The decisive variable is *when*, not *what* or *how much*.
3. **Parity** (§4): guards trade repair for dose, and the guarded pulse at dose 2.0 statistically ties text interruption on judged surprise with identical connection — replicating in-family at 14B with an exact tie (§5).
4. **A mechanism** (§6): habituation leaves a silent schematic orbit, measurable as matched token–state periodicity; interruption and the pulse both flatten it.
5. **A dissociation** (§7): on verified code search the same operators buy zero tail depth — the state channel moves judged experience, not discovered value.

Every confirmatory analysis below was pre-registered (grid, primary contrast, guards, verdict tiers) and written as code before the data existed; exploratory screens are labeled as such. All deltas are paired per premise over the ten held-out premises of Ono Filho [11]’s confirmatory battery, with exact sign-flip tests (2^{10}) and bootstrap 95% CIs.

2 Instrument and protocol

Operators. All operators are rank-1/2 updates from stored unit vectors, $O(d)$ per position; no $d \times d$ matrix is ever materialized. With $\hat{u}, \hat{w}$ orthonormal and $c_u = \langle h, \hat{u} \rangle$: *translation* $h + \alpha \hat{v}$; *rotor* $h + \sin \theta (c_u \hat{w} - c_w \hat{u}) + (\cos \theta - 1)(c_u \hat{u} + c_w \hat{w})$, the closed (Rodrigues) form of $\exp(\theta(\hat{w}\hat{u}^\top - \hat{u}\hat{w}^\top))$, verified against the matrix exponential to 10^{-10} ; *projection/dilation* of a subspace; *repulsion* $h + \beta (h - \mu) / \|h - \mu\|$ from an exponential moving average μ of the trajectory. Two composable *guards*: an anchor that restores the coordinate along the premise direction after the operator, and a norm shell that rescales $\|h\|$ to its pre-operator value.

Directions and dose. Concept directions are the model’s own: the unit mean hidden state of a concept text at the target layer (position 0 excluded), differenced against the premise state. Doses are expressed in units of the premise-state norm at that layer, making α dimensionless and, as §5 shows, transferable across models without retuning.

Carrier and measurement. Unless noted, generation runs on a base model (Qwen3-8B) with the habituation of Ono Filho [11] (a count-weighted log-penalty over the last 512 tokens) — the regime in which text interruption was established. Judging follows the generated-only protocol: 96-token windows that never contain injected text, scored on surprise, connection and coherence by a $k=5$ judge panel whose test–retest behavior was measured in the original study. A cheap coherence proxy (mean NLL of the stream under the same model, untouched) and distinct-4 accompany every cell. Verified experiments (§7) reuse the exact bin-packing evaluator and classical baselines of Ono Filho [12].

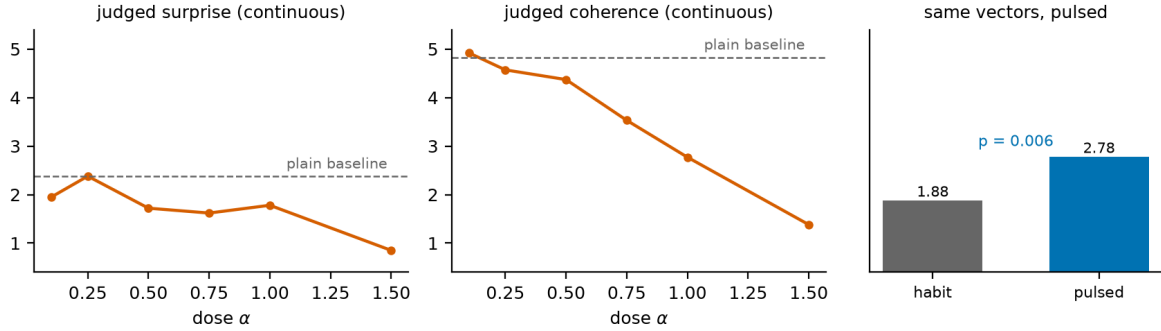


Figure 1: The temporal law. Left, center: judged surprise and coherence at layer 18 as continuous dose rises, against the plain-sampling baseline of that battery (dashed) — more push, less surprise, collapsing coherence. Right: the same directions delivered as 32-token pulses every 300 on the habituated carrier, against that carrier’s own baseline (+0.90, $p = 0.006$). The vectors are identical; only the temporal regime differs.

3 The temporal law

Continuous injection has no fertile band. An exploratory depth $\times$ dose map (57 cells) located the semantic lever in the middle of the network: early-layer injection collapses generation into memorized register basins (a math-homework attractor), late-layer injection acts as token bias — at high dose the concept’s own token fragment becomes a literal loop — and only mid-network injection transports content. The pre-registered confirmatory then swept the candidate band (layers 14/18/22, $\alpha \in [0.1, 1.5]$, continuous schedule, 140 cells) with judges. The primary failed in both halves: band-core surprise minus the habituated baseline $-0.478 [-1.478, +0.522]$, one-sided $p = 0.81$, and the coherence guard failed (-1.08). The dose-response is monotone the wrong way: surprise $2.38 \rightarrow 0.85$ and coherence $4.83 \rightarrow 1.38$ as α rises to 1.5 (Fig. 1). More push makes text that a judge finds *less* surprising: degeneration is boring. Notably, the prompt-matched control — the same concept injected as text into this plain carrier — also failed to beat the baseline, reproducing the original finding that novelty injection needs habituation as its carrier [11].

The same vectors, pulsed, are fertile. The pre-registered transposition of the interruption’s temporal structure — bursts of 32 tokens every 300, directions rotating over the same four new-subject texts used by text interruption, on the habituated carrier — raises judged surprise by $+0.90 [+0.45, +1.33]$ over habituation ($p = 0.0059$, coherence guard passed). Same directions, same layer, same per-token dose as a continuous regime that failed at $p = 0.81$: the decisive variable is the temporal regime. Text interruption replicates in the same battery ($+1.76$, $p = 0.0029$), anchoring the instrument.

4 Guards license dose; parity with the written word

Pulsing at higher dose without protection costs coherence (-1.01 against text interruption, $p = 0.049$). The pre-registered guard test shows the guards do *not* repair this: at equal dose, anchor + norm-shell change coherence by $+0.02$ ($p = 0.50$) while visibly weakening the effective push (clean NLL $2.53 \rightarrow 1.83$). What they do is license dose: under guards the ladder $\alpha = 1.0 \rightarrow 1.5 \rightarrow 2.0$ climbs monotonically in judged surprise ($2.10 \rightarrow 3.23 \rightarrow 3.40$) at roughly constant coherence (Fig. 2).

At the top of that ladder sits the pre-registered comparison against the program’s strongest prompt-side intervention. The verdict tier that fired is the middle one: **parity**. Guarded

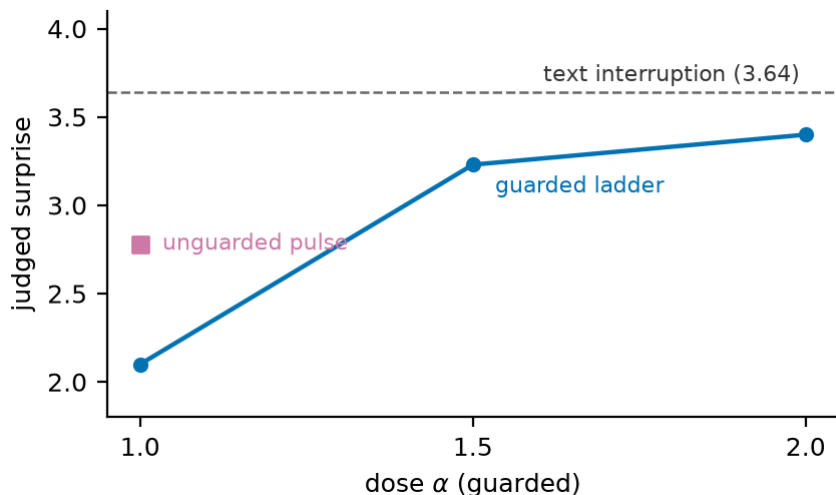


Figure 2: Guards license dose. Judged surprise of the guarded pulse at doses 1.0/1.5/2.0 (points), with the unguarded pulse (square) and the text-interruption reference (dashed line). The ladder reaches parity.

pulse at dose 2.0 versus text interruption: surprise 3.40 vs 3.64 ($\Delta = -0.24$ $[-0.94, +0.47]$, two-sided $p = 0.53$, $|\Delta| < 0.5$), connection 2.38 vs 2.39, coherence within the pre-registered guard. A subject can be placed into the model’s stream as pure activation — invisible in the text, spending no tokens, continuously dosable — and a judge panel cannot tell the difference from writing it down. We emphasize what parity is not: the strong hypothesis, that state injection reaches value no prompt reaches, remains open (§10).

Two declared secondaries returned honest negatives. A first-version rotor (rotating the premise–subject plane, $\theta = 0.8$) underperforms text interruption (-1.53 , $p = 0.014$; a later cheap screen located its useful range at $\theta \in [1.2, 1.6]$ — an untuned operator, not a dead one). And the program’s own registered prediction failed: pulsed state repulsion *without* an anchor did not degenerate (its coherence was the highest of the state arms), refusing the fourth appearance of the anchor law; a stress screen then showed why — repulsion from a *moving* mean is self-limiting, because the reference chases the state. The anchor law, as established in weights and behavior, is about fixed references.

5 Generality: the parity travels in-family

The pre-registered generality battery reran the decisive trio — habituated baseline, guarded pulse at dose 2.0, text interruption — on three untouched models, with the layer at 50% of each network’s depth and the dose carried over unchanged in premise-norm units.

Qwen3-14B (same family, larger): full replication. Pulse over habituation $+1.083$ $[+0.567, +1.617]$, $p = 0.0020$, coherence guard clean; and parity with text interruption lands as an exact tie: surprise 3.48 vs 3.48 ($p = 0.99$). The dimensionless dose and proportional-depth layer transferred without retuning.

Qwen3.8-27B (current generation, aligned): the carrier breaks, not the phenomenon. Under our base-model protocol — which bans EOS and forces 1500 tokens — this chat-trained model opens with fluent prose and collapses into token salad past its natural turn end (judge notes: “degenerate token soup”). A nominally significant pulse effect inside that salad ($p = 0.041$) does not count as support, and we do not count it. Aligned carriers need their own protocol (chat-formatted seeding, free EOS, natural lengths); this is future work, not a footnote.

model	pulse – habit (primary)	pulse – inter	verdict
Qwen3-8B (origin)	+0.90, $p = 0.006$	−0.24, $p = 0.53$	parity
Qwen3-14B	+1.08, $p = 0.002$	+0.005, $p = 0.99$	parity (exact tie)
Qwen3.8-27B (aligned)	carrier breaks under forced continuation		no reading
OLMo-2-13B	−0.09, $p = 0.78$	—	null (register basin)

Table 1: The generality ladder. Judged-surprise contrasts, paired over the ten premises; doses in premise-norm units and layers at 50% depth, carried over without retuning.

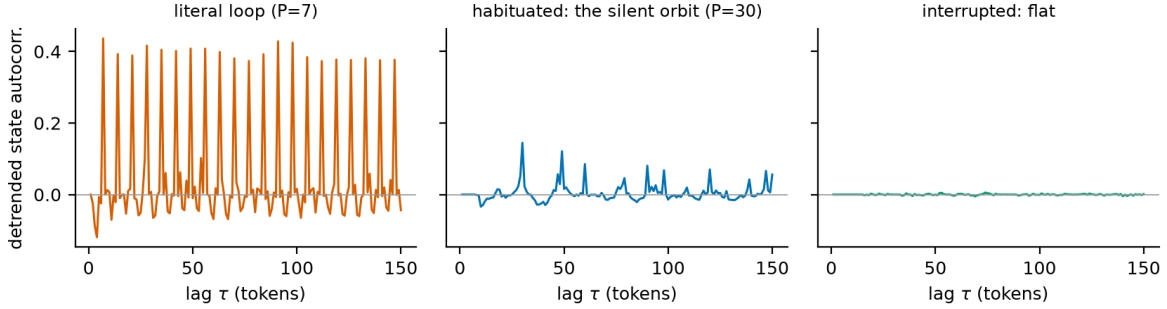


Figure 3: Silent orbits and their removal. Detrended state autocorrelation for a literal loop (state period = phrase length), the habituated carrier (the silent orbit at $P \approx 30$, matched in tokens), and an interrupted cell (flat).

OLMo-2-13B (different family): a real null with a carrier caveat. Pulse minus habituation −0.09 ($p = 0.78$). All three arms, however, sit in an encyclopedic register basin (judged coherence 0.4–0.6 everywhere; heavy judge-refusal rate), the OLMo analogue of the register basins of §3. Whether the null reflects family-dependence of the phenomenon or miscalibration of layer and dose for this family cannot be separated here; per-model calibration is registered future work. The generality claim we make is therefore exactly this: *the parity travels across scale within a family; family-generality is open.*

6 The mechanism: silent orbits, flattened spectra

Why does the pulse work where pressure fails? A measurement study replayed finished cells through the model, capturing the mid-layer trajectory, and compared the state autocorrelation $A_{\text{state}}(\tau)$ (unit-normed, mean-centered states) against the token autocorrelation (the ground truth of literal loops), reading the strongest detrended local peak of each ($\tau \geq 5$; literal period-2 loops sit below the reader’s floor and are noted as a blind spot).

The prominence ladder is clean (Fig. 3). A literal phrase loop shows a state period matching the phrase length ($P = 7$, prominence 0.44). Register collapses — the quiz mode, the math-exercise mode — show matched token–state periodicity at their block rhythms ($P = 123$ and 84). And the habituated carrier, which repeats no tokens, still carries a *silent schematic orbit*: matched token–state periodicity at $P \approx 30$ (prominence 0.18/0.15) whose text reads as sentence-level cycling through the same cast — the “return to the theme, not the sentence” of Ono Filho [11], now visible as a rhythm in the state. Interruption, the pulse, and state repulsion all flatten this internal spectrum (prominence ≤ 0.014). The interventions that judges reward are, mechanically, spectral flatteners of the thought trajectory.

7 The verified dissociation

Do these operators buy anything a verifier can price? The pre-registered battery ran three arms — untouched carrier, pulses toward narrative subjects (the cross-register wildcard), pulses toward the domain’s own thinking prompts — inside verified bin-packing notebook generation (16 distribution variants including the far family of Ono Filho [12], 6 notebooks each, 288 notebooks), scoring with the exact evaluator against best-fit and first-fit.

The answer is a clean null with a striking signature. Per-variant best distance to the classics: wildcard minus control $+0.00014$ [$+0.00000, +0.00036$], one-sided $p = 1.0$; transfer and far-family contrasts null; and the single best gap across *all three arms* is identical to five decimals (-0.00334 , three times). At this scale the reachable tail depth behaves as a constant of the model and carrier, untouched by state stimulation. (An earlier 120-notebook screen had produced a lone wildcard find; the confirmatory dissolved it into $n=3$ sampling noise — all arms find occasionally at $n=6$ — which is the two-gear system doing its job.) A first-version route-blocking operator — projecting out the top principal directions of a previous solution’s trajectory — behaved as a mild handicap: production down, no depth gained, and a calibration curve showing that the top trajectory components carry the code-writing machinery itself (blocking ≥ 4 directions is lobotomy).

Together with §4, this is the paper’s deepest result: *the state channel moves judged experience, not discovered value*. The operators that tie the written word in a judge’s eyes buy zero verified tail — the oldest lesson of the program (surface novelty is not value), returned at the deepest substrate, and consistent with the attractor results of Ono Filho [12].

8 Boundaries

For designers of state interventions, the mapped edges: (i) *depth matters more than dose* — early layers destabilize into register basins, late layers are token bias (at high dose the injected concept’s token fragment loops literally: the late network is a mouth, not a mind); (ii) *guards are throttles, not medicine*; (iii) *moving-reference repulsion is self-limiting* — collapse requires a fixed reference, which reframes the anchor law; (iv) *aligned models are a different carrier* — forced continuation past the turn end manufactures degeneration that no operator caused; (v) judged-surprise instruments punish degeneration (they cannot be gamed by noise), but they refuse some degenerate text outright, which costs coverage and must be reported.

9 Related work

Activation steering. Additive steering vectors [13, 16], truthfulness intervention [6], representation engineering [18], SAE features [3], function vectors [15] and refusal-direction ablation [1] steer style, safety and factual behavior; industrial agents already use such edits routinely for control [2]. STARS [17] is closest in spirit — multiple orthogonal steering directions for diverse generation — but evaluates without a prompt-matched rival or a deception-resistant judge protocol, and uses translations only. Mishra et al. [10] prove steered states are unreachable by prompts but do not test their value; our parity-not-superiority result is the first value measurement in that opened space, and it constrains easy versions of the fertility hypothesis. Weight-space analogues — task arithmetic [5], creativity weight-steering [14] — edit the persistent model where we perturb the transient thought.

Mechanistic intervention. Causal tracing and model editing record and modify internal routes to *explain* behavior [9]; our route-blocking screen inverts the direction — recording a

route to *forbid* it — and reports the first calibration of how much of a trajectory can be blocked before the faculty, not the idea, is removed.

Decoding-time control. Degeneration under maximization [4] anticipates our salad regimes; contrastive decoding [7] and proxy tuning [8] intervene through logits of a second model, a surface channel our late-layer results effectively reproduce (late-layer injection $\approx$ token bias); the mid-network channel studied here is precisely what logit-level methods cannot reach.

The program. Protocol, habituation and interruption from Ono Filho [11]; consolidation, tails and the attractor from Ono Filho [12]; the operator-package factorial from the companion verified-search paper. The AAR study [2] independently reproduces the program’s telemetry (method monocultures, flat idea diversity under climbing scores, capability anchors) at industrial scale in an alignment-mitigation setting — a mass regime, complementary to the frontier questions here.

10 Discussion and limitations

What parity means. A new channel into the model’s ongoing thought exists, is dosable and reversible, spends no context, is invisible in the text, and matches the best prompt-side intervention under a judge panel that cannot see either intervention. What it does not mean: superiority. The strong fertility hypothesis — that prompt-unreachable states yield value prompts cannot — remains open; our continuous-injection null and the state $\approx$ prompt tie in the easy setting are its first constraining data points.

Experience versus value. The dissociation is the finding we expect to age best: state operators changed what a judge experiences and left untouched what a verifier prices. Any proposal that inference-time steering will unlock discovery must now say why its mechanism differs from the ones measured here.

Limitations. The ten premises are reused across this paper’s batteries (each analysis pre-registered before its data, but adaptation risk accumulates and is declared); the judge panel is a single instrument family (its test–retest behavior was measured in 11); generality is established across scale, not family; doses and layers were calibrated on the origin model and transferred by convention (50% depth, premise-norm units) rather than per-model tuning; and the verified domain is one task family. The temporal law’s mechanism (why 32-token pulses integrate where constant pressure degrades) is characterized spectrally but not causally.

Future work. A protocol for aligned carriers; per-model calibration for cross-family tests; the strong fertility test on a carrier where the pulse and the word are both at ceiling; and substrate-level questions — consolidating the search’s schematic memory rather than its winners, and cross-model state transfer — that the present results motivate but do not touch.

Reproducibility

The operator algebra, both backends, every generation and judging script, all cells, judge verdicts, and the dated pre-registrations are released at <https://github.com/RobertoOno/interrupting-the-loop>. All confirmatory analyses were committed as code before their data existed.

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